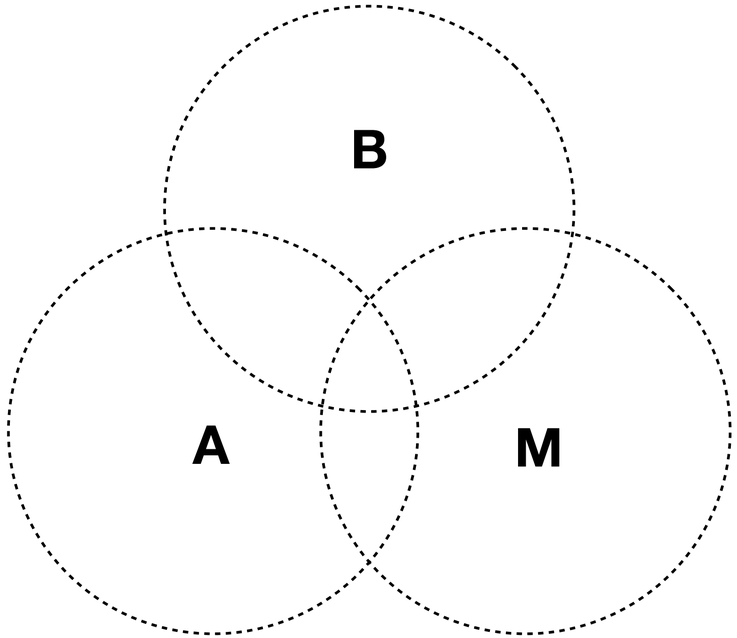
Session 1: Foundations of Marine Macroecology and Niche Theory

Modeling the distribution of biodiversity under global climate change

Jorge Assis, PhD // Nord University, Norway // biodiversityDS, Centre of Marine Sciences, Portugal

Connecting Niche to Geography



Connecting niche to geographic space: BAM diagram

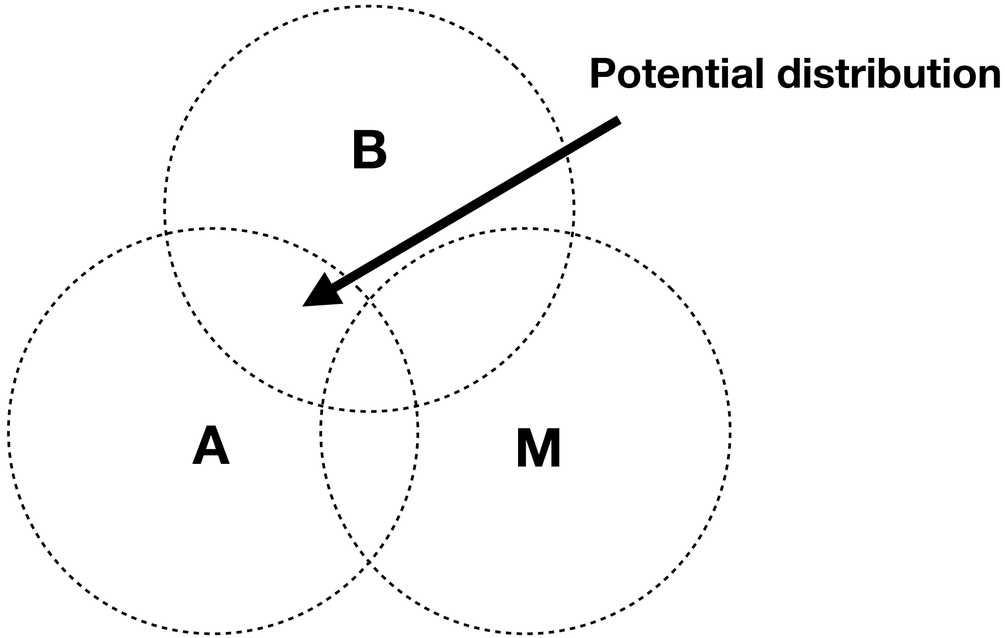
A conceptual model translating niche theory into geographic space. Highlights **three intersecting factors determining species' distributions:**

Biotic: suitable biotic conditions (presence of prey, absence of strong competitors).

Abiotic: suitable environmental conditions (temperature, salinity, depth, etc.).

Movement: accessibility – the regions reachable by the species via dispersal.

Soberón & Peterson, 2005

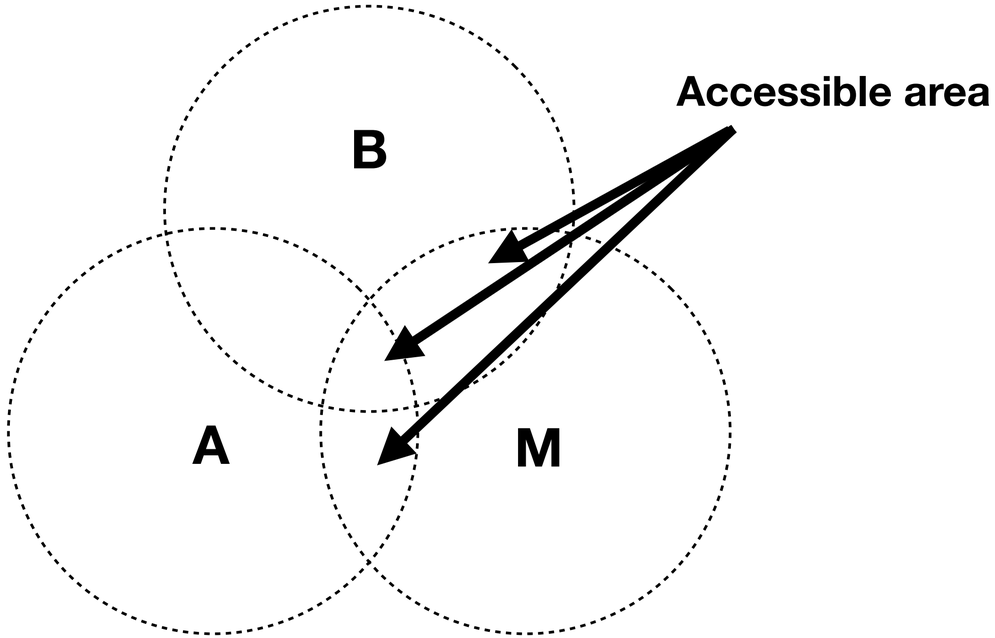


Potential distribution (A ∩ B): The geographic area where both **abiotic (A) and biotic (B) conditions are suitable** - link to the realized rich of Hutchinson.

Accessible area (M): This is the **geographic area the species can actually reach through dispersal** within a given time frame.

Occupied distribution (A ∩ B ∩ M): The **geographic area where the species is actually found**. It's the part of the potential distribution that is also accessible.

Soberón & Peterson, 2005

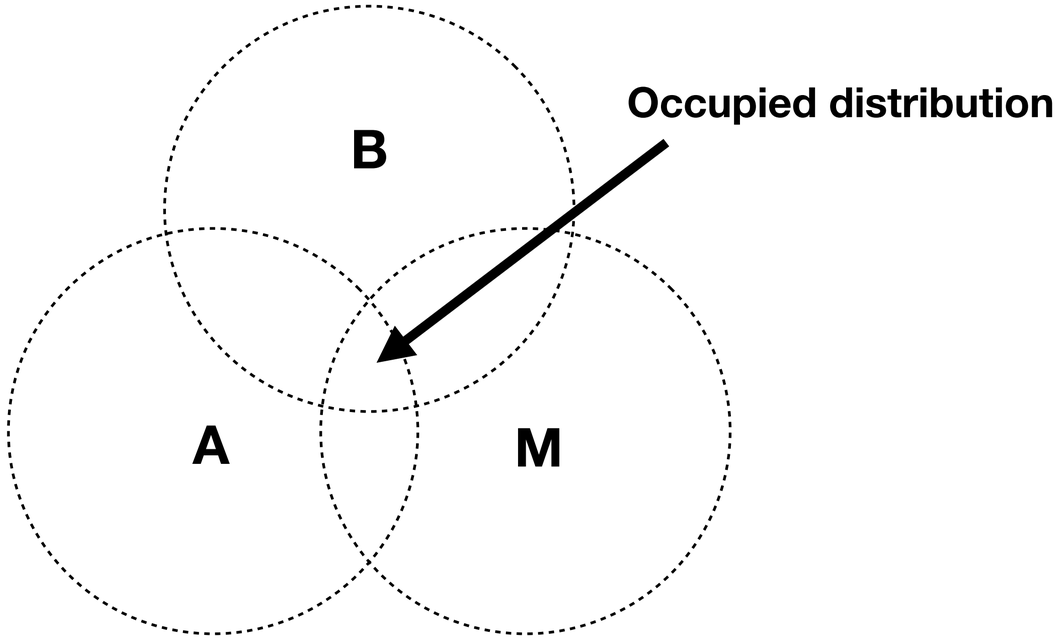


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Soberón & Peterson, 2005

2 min. PITCH

Papers: P30, P38

10 minutes

Read the abstract and the introduction and interpret the main figures.
(P30 Figs 1 and 2; P38 Figs 1 and 3).

Question. What problem did the authors tackle? (one sentence)
Approach. What did they do, and with which data and model? (one sentence)
Key finding. The one result to remember, ideally with the paper's main figure.
Why it matters today?